

Ishaan Pathak

Southfield, Michigan (Open to relocate) | ishaan175pathak@gmail.com | [LinkedIn](#) | [Github](#) | [Researchgate](#)

EDUCATION

Lawrence Technological University

Master's of Science in Computer Science

Teaching Assistant: Python & Robotics

Southfield, MI

August 2024 – July 2026

Devi Ahilya VishwaVidyalaya

Bachelor of Computer Application (Honours)

Indore, India

August 2019 – July 2023

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, HTML, CSS

AI/ML: PyTorch, TensorFlow, Keras, Transformers (Hugging Face), Computer Vision, RNNs, CNNs, YOLO, scikit-learn, scipy

Generative AI: LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, Semantic Search

Web Development: React.js, Node.js, Express.js, Flask, Django, Spring Boot, Three.js, REST APIs

Cloud & DevOps: AWS (Lambda, DynamoDB, S3 Buckets), Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Kafka

Databases: MySQL, PostgreSQL, MongoDB, Oracle, Redis, NoSQL, GraphQL

EXPERIENCE

Research Assistant

Lawrence Technological University

February 2025 – July 2026

Southfield, MI

- **Achieved 81% mAP** on YOLO-based crack detection models by curating and **annotating a 10,000+** image dataset
- **Developed** state-of-the-art Computer Vision models for medical detection, contributing to automated medical research.
- **Assisted** professor in **reviewing, refining, publishing** research manuscripts for clarity, rigor, and publication readiness.

Student Web Developer

Lawrence Technological University

May 2025 – August 2025

Southfield, MI

- **Developed** Web app for anonymous essay submissions, enabling over **50+ students** to participate in bias-free evaluations.
- **Integrated** a Gen AI model to benchmark automated grading against human assessments, **improving** consistency by **25%**.
- **Optimized** the grading workflow, **reducing** manual evaluation time by **40%** and grading throughput **increased by 35%**.

Machine Learning Engineer

GenieTalk.ai

June 2022 – February 2023

Indore, India

- **Developed** and **optimized** machine learning models for text classification using TensorFlow and scikit-learn.
- **Enhanced** algorithm performance and **integrated** AWS for scalable model deployment.
- **Developed** Application Programming Interface (APIs) for model integration, enhancing response times by **20%**.

PROJECTS

MedBot360 – AI Health Companion | MERN, Transformers (Hugging Face), Docker, AWS

- **Built** a MERN-based health dashboard enabling secure symptom tracking and personalized recommendations for **200+** users.
- **Integrated** a Hugging Face-powered medical chatbot for real-time symptom analysis, boosting accuracy by **30%**
- Containerized the full stack using **Docker** and deployed on **AWS** for high availability and scalability.

FinWise – AI Financial Advisor Dashboard | MERN, Transformers (Hugging Face), Redis, LangChain, AWS

- **Designed** GPT-powered financial agent analyzing Plaid data for personalized insights, improving decisions by **40%**.
- **Built** React + Chart.js dashboard with Redis caching for low-latency financial analytics.
- **Enabled** natural language queries using LangChain retrieval for instant financial advice.

Social Media Analysis | Apache Spark, Scikit-Learn, Python (Matplotlib, Seaborn, Pandas), SQL

- **Engineered** an **Apache Spark** pipeline to process high-volume tweet data using distributed computing.
- **Implemented K-Means** and **PCA** to automate content segmentation and identify sentiment trends.
- **Automated** visual reporting via **Matplotlib/Seaborn** to deliver actionable insights for brand monitoring.

Personalized Search Reranker | BERT, Cross-Encoders, TREC DL, PyTorch

- **Boosted** search accuracy (**MRR@10 to 0.0406**) by developing a reranking pipeline integrated with clickstream data.
- Predicted **query-document relevance** by fine-tuning Sentence-BERT on MS MARCO using binary cross-entropy.
- Enhanced **inference efficiency by 15%** by scaling Python preprocessing and optimizing **AdamW** training.

RESEARCH, AWARDS & CERTIFICATIONS

- Publications: **AI-Powered Music Generation from Sequential Motion Signals** (IJIS, 2025); **Advancing Automated Brain Tumor Detection via YOLOv11** (JCTR, 2025)
- Research: **Advanced Smart Infrastructure Damage Detection** utilizing Computer Vision and AI for automated structural health monitoring and defect detection.
- **Outstanding Research Assistant in Data Science Award** – Lawrence Technological University (awarded for technical excellence and research contribution).